

SPECTRA — brief for Gilad

Ido Paretsky · advisor Dr. Gilad Katz · BGU · 16 Aug 2026. Same idea as NEON (Hirsch and Katz, 2022): one offline DRL agent, many architectures, user-set accuracy preference τ . SPECTRA is that idea for structured CNN pruning. Paper due 30 Sep. Test-set Δacc unless noted. Eval quotes at $\geq 70\%$ params kept.

Status toward the 30 Sep paper

On CIFAR-10, one offline DRL agent — trained once on a catalog of CNNs — prunes unseen networks (ResNet, VGG, MobileNet, DenseNet) with no extra agent training. Test-set accuracy drop (Δacc) is a few percentage points on several held-out checkpoints. Best transfer so far: train on CIFAR-10 ResNets, then prune a thin ResNet-20 the agent never saw — Δacc -3.1 percentage points, 60% of parameters and 79% of FLOPs kept. The hardest held-out case so far (thin ResNet-56, width 4) is Δacc -15.9 percentage points at 70% parameters; the plan to move it is a larger train catalog (24 nets, already running) and a size-matched greedy control. Experiment freeze 15 Sep, paper 30 Sep.

What we have

A reusable A2C policy for structured CNN pruning. Train once; apply to a new checkpoint with no extra agent training. The user sets an allowed accuracy drop τ (we use 10 percentage points). Similar-family TEST of the 10-net agent is in: same families as training, new widths and sources.

What we just measured

10-net catalog (CIFAR-10, SVHN, Fashion-MNIST). Similar-family TEST: DenseNet-100 Δacc -2.0 pp at 85% params. Held-out thin ResNet-56: Δacc -15.9 pp. Greedy / random rate-pickers now run in the same prune-and-fine-tune loop so later tables can show the learned schedule against a control.

Next milestones (on the cluster)

Unlike-family eval (ShuffleNet, RepVGG) is running. A 24-net train catalog is running — test whether the thin ResNet-56 moves. Then: greedy at the same parameter ratio as the agent. CIFAR-100 fine-tune recipe is in progress; DRL on that dataset starts once a real mid-layer cut holds accuracy.

Road to the end-game product

By 15 Sep freeze the agent, catalogs, and tables. By 30 Sep the paper is one transferable structured-CNN pruning policy under a user-set accuracy budget, with held-out eval and greedy/random controls. Not ImageNet SOTA. Not training on all 287 checkpoints — genericity is held-out eval.

–3.1 pp

Train C10 ResNets $\rightarrow$ unseen thin ResNet-20, keep 60% params

–15.9 pp

Held-out thin ResNet-56; next lever is a larger catalog

–2.0 pp

10-net agent, similar-family DenseNet-100 TEST, 85% params

30 Sep

Paper due (experiment freeze 15 Sep)

What SPECTRA is, and the end-game

Generic (from NEON)

Train the agent once on many CNNs. Prune a new net with no extra agent training. Not AMC-style per-network RL.

Preference (from NEON)

User sets allowed accuracy drop τ (we use 10 percentage points). Compress, but do not cross that line. Same reward trichotomy.

CNN, structured (new)

Remove channels so shapes change. Skip connections and shuffles shrink as groups. Fine-tune after each prune. Small Transformer reads the net; frozen BERT is an ablation, not the default.

How to read the numbers

Why 1.0 / 0.9 / 0.8, not NEON's 0.7 / 0.6

NEON's legal cuts were keep 100/90/80/70/60% of a dense layer. SPECTRA's default is keep 100/90/80%. A CNN action resizes a residual group, and rates compound across layers. Without a size floor, deeper cuts on a thin ResNet-20 reached ~40% params at $\Delta\text{acc} = 12$ pp — already past $\tau=10$ — and unconstrained eval hit ~17% params at about -40 pp. Greedy 0.8 alone is already $\Delta\text{acc} = 24$ pp on the held-out thin ResNet-56. 0.7/0.6 would be a one-shot path past a recoverable size; they stay out of the default menu.

Who picks the rate vs the filters

At each layer the environment always drops the lowest-magnitude channels (Li et al. 2017). The policy only chooses how hard to cut that layer: keep all / cut 10% / cut 20%. Learned DRL, greedy, mild, and random are four rate-pickers in the same prune, fine-tune, and 70% size-floor loop.

What "beats greedy" means

Greedy always takes the strongest legal cut (keep 80%). The logs historically called this "L1"; that name is the rate heuristic, not a different ranking. Random samples a legal cut. Mild prefers a lighter cut. On the easy net they tie; on the thin ResNet-56 Δacc is 8 percentage points better.

Policy	What it chooses	Thin ResNet-20 (easy)	Thin ResNet-56 (hard, held out)
Learned DRL	A2C actor: which layers, how hard	-4.2 pp at 60% params	-15.9 pp at 70% params / 55% FLOPs
Greedy (log name: L1)	Always the strongest legal cut (keep 80%)	-4.3 pp at 60% params	-24.0 pp at 67% params / 45% FLOPs
Mild	Prefer keep 90% if legal	-4.1 pp at 60% params	-24.1 pp at 67% params
Random	Uniform among legal non-identity cuts	-5.0 pp at 60% params	-24.1 pp at 67% params

Test set, CIFAR-10, $\geq 70\%$ params kept as the quoting floor. Same 40-epoch fine-tune. Filters are always ranked by magnitude; only the rate schedule differs. DRL keeps more of the hard net (70% vs ~67%), so a matched-size greedy run is still owed. "Thin" = CIFAR ResNet with few channels per layer (width 2 or 4); held-out = not in the train catalog. Source: jobs 20189046 (DRL), 20189043 (greedy), 20189044 (mild), 20189045 (random).

Desired end-game (the paper)

One offline DRL agent that prunes unseen structured CNNs under a user-set accuracy budget τ , with tables that mean what they say: train catalog vs held-out checkpoint, similar vs unlike families, learned schedule vs greedy/random rate-pickers. CIFAR-100 joins the train set once the fine-tune recipe supports a real mid-layer cut. Not DepGraph on ImageNet. Not training on all 287 checkpoints — genericity is held-out eval.

TOP-5 code changes since 8 Aug (overhaul of the inherited repo)

Starting point: commit ecefe78. ~15.6k lines added since. These are the changes that made SPECTRA a real experiment, not the later seed replicates.

#	What changed	Why it matters
1	The checkout could not run, and compression was fake	Entry point and utils disagreed (AttributeError). torch.nn prune zeroed weights but left shapes, so param/FLOP counts never fell; a second prune re-selected already-zero filters. The resize path assigned into a Python list and never rebound the module — and would have thrown away pretrained weights. Fine-tune was for epoch in range(0), then infinite reinit recursion. Episode return stored the last step, not the trajectory. Entropy was computed and discarded. Nested DDP wrapped the CNN every step (7 s → 21 s). Dataset cache keyed on the wrong dict, so hundreds of nets each spawned 3×8 persistent workers. Fixed: single-GPU default, structural rebuild of layers, deepcopy of the baseline, DatasetRegistry, real fine-tune, usable entropy, correct returns.
2	Structured prune that respects CNN connectivity	torch.fx channel groups: residual adds, DenseNet concat offsets, ShuffleNet, depthwise ties. Coupled layers drop the same indices. Masked zeros get no compression credit in the reward. Classifier output is never shrunk. Verified: VGG/ResNet/DenseNet/MobileNet actually lose 33–58% params at a uniform 0.8 pass.
3	State encoder: small Transformer trained with A2C, not frozen BERT	NEON's fixed maps do not fit variable-depth CNNs with skips. Frozen bert-base-uncased (~110M, English) cannot do representation learning for filter norms. Default is a ~2–3M Transformer: one token per layer, learned target marker, attention bias on coupling ids, action-cost features, database-wide z-score. BERT remains an ablation. Encoder capacity did not fix the thin deep ResNet-56 — catalog diversity did.
4	Preference-aware protocol and honest baselines	Stem and width-1 layers cannot take fake prunes. Eval identity-pads once parameters kept hit 70% — otherwise thin nets go to ~17% size and –40 pp, which is a different question. Training still uses τ , not that floor. Greedy / mild / random pick rates in the same prune + fine-tune + floor loop, so DRL numbers are comparable.
5	Train / similar / novel catalogs and unattended cluster runs	287 checkpoints are mapped; the agent trains on a subset. Similar = same families, different width/depth/source. Novel = families never in train (ShuffleNet, RepVGG). CIFAR-10 jobs no longer silently load CIFAR-100. Training stops on a timer; evaluation still runs. Successor jobs start only if the parent exits 0. 7-day wall so a slow DenseNet eval is not killed.

TOP-5 insights

#	Insight	Consequence
1	A prune action that did not change width was still scored as an RL step.	On thin ResNets (6–16 channels) $\text{ceil}(0.9 \times 6) = 6$: the environment applied nothing while A2C still got a reward or penalty. Any requested compression now drops at least one channel. A 0.8 rate is also not a 20% size cut — residual groups decide params/FLOPs, so tables quote those, not the menu.
2	Until the overhaul, reported size cuts could not have been real.	Every later table rests on shapes that actually change and rewards that match those shapes.
3	CIFAR-10 is a recoverable training environment. CIFAR-100, under Adam 40–80 epoch fine-tune, is not.	Do not mix CIFAR-100 into the agent until a 2–5% structured cut recovers inside –10 pp. That is fine-tune and data, not “the Transformer is too small.” We are now testing SGD + cosine + MixUp / distillation at 160 epochs.
4	More diverse training nets moved the thin ResNet-56. Encoder size did not.	Set-encoder, wider Transformer, and frozen BERT all left that net near –24 pp. Training on ten C10/SVHN/Fashion-MNIST families moved it to –15.9 pp. Putting one thin net in train did not. A 24-net catalog of the same cheap datasets is running now.
5	The learned schedule only beats greedy on the hard net. Easy nets are a tie.	Thin ResNet-20: DRL and greedy both about –4 to –5 pp at 60% params. The generic-agent claim is the hard transfer, not the easy Pareto.

TOP-5 results (test, ≥70% params kept unless noted)

#	Result	Read as
1	Train on CIFAR-10 ResNets; test on a thin ResNet-20 never in that catalog: Δacc -3.1 pp at 60% params / 79% FLOPs (CIFAR-10).	Policy transfer: no extra agent training on the eval net. Best easy-net Pareto so far.
2	Thin 56-layer ResNet (width 4), held out: Δacc -15.9 pp at 70% params. Plan: 24-net catalog (running) and greedy at the same size.	Still misses $\tau = 10$ percentage points. Greedy in the same loop is -24.0 pp at 67% params — a matched-size greedy run is the fairness check, not the headline.
3	DenseNet-100 -2.1 pp at 83% params; ShuffleNet- $v2 \times 1$ -1.7 pp at 88%. Same agent's thin ResNet-56 was still -24.5 pp.	Other families work when they are in (or near) the training set. Transfer is not uniform.
4	Same families, different width/source: ResNet-56 width 10 -9.2 pp; VGG-19 BN -2.7 ; MobileNet- $v2 \times 0.75$ -2.5 ; DenseNet-100 -2.0 pp at 85% params / 85% FLOPs.	In-distribution generalizability, not the thin deep ResNet-56 stress case. Unlike-family (ShuffleNet, RepVGG) eval is now running.
5	CIFAR-100, old fine-tune (Adam, 40–80 epochs): no 2–5% channel cut returned within -10 pp of original test accuracy. The one crop+flip 'success' still kept 99.5% of the parameters.	Not an action menu. New probes (SGD, cosine, MixUp, AutoAugment, 160 epochs; plus knowledge distillation) are running on competent CIFAR-100 nets. DRL on CIFAR-100 starts only if a real 1–5% cut lands inside -10 pp.

Next milestones → 30 Sep

15 Aug → 30 Sep

31 experiment days, then 15 writing days

Orange: finish held-out evals already on the cluster. Blue: larger catalog, CIFAR-100 recoverability, matched-size greedy. Green: write-only after 15 Sep. Arena = generic offline DRL on structured CNNs with preference τ , vs greedy/random rate-pickers.

TOP-5 open points

#	Open	Stance
1	Thin ResNet-56 (width 4) is still -15.9 pp ($\tau=10$ missed). Train split was only -1.7 pp.	Best so far. Larger catalog is the next lever. Matched-size greedy still needed.
2	CIFAR-100 2–5% structured cuts do not recover under the old fine-tune.	Not given up. New fine-tune is on the cluster. Negative only if that also fails.
3	Unlike-family eval (ShuffleNet, RepVGG) is not in yet.	Queued behind the 10-net agent's similar eval. Do not rewrite the story until those numbers exist.
4	287-file pool vs a 10- or 24-net train set.	Cover the pool by evaluating held-out CNNs, not by training on every file. Drop ViT/ImageNet-in-the-loop, grafting toys, and under-trained thin nets.
5	Fifteen writing days after 15 Sep. ImageNet / ViT / DepGraph SOTA will not fit.	Freeze the arena. Thesis = NEON transferred to structured CNNs, honest baselines, known failure modes.

Defaults unless a later number wins

Eval floor 0.70. Small Transformer (not BERT). NEON reward. Illegal-action mask on. Rates 1.0 / 0.9 / 0.8 (NEON's 0.7 / 0.6 omitted — see above). $\tau=10$. Full-net fine-tune, 40 epochs, patience 10. AMP off. CIFAR-100 out of the DRL train set until a 2–5% cut recovers.

NEON → structured CNN

Measurements are real

Thin R56: DRL -15.9 vs greedy -24

Freeze 15 Sep · paper 30 Sep